1. Prove the following *generalized Liouville's theorem*: If f is an entire function and $|f(z)| \leq C(1+|z|^N)$ for all $z \in \mathbb{C}$, then f is a polynomial of degree at most N.

2. Prove the following *Cartan's uniqueness theorem*. Let f be a holomorphic self-mapping of a bounded domain Ω. Suppose that there exists $z_0 \in \Omega$ such that $f(z_0) = z_0$ and $f'(z_0) = 1$. Then f is the identity.
Hint. Without loss of generality, suppose that $z_0 = 0$. Assume that $f(z) = z + a_k z^k + O(z^{k+1})$ with $a_k \neq 0$ for some $k \geq 2$. Denote by f^n the n-th iterate of f. Apply the Cauchy estimates to f^n to get a contradiction.

3. Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be a power series with radius of convergence $\rho > 0$. Prove the following *Gutzmer's inequality*:

$$\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq A_f(r)^2, \quad 0 < r < \rho,$$

where $A_f(r) := \max\{|f(z)| : \ |z| = r\}$.
Hint. Consider the integral $\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta$.

4. Compute

$$\oint_{|z|=1} \frac{|dz|}{|z-a|^2}, \quad 0 < |a| < 1,$$

by transforming the integral into a form where you can apply the Cauchy integral formula.

5. Prove the following version of the *mean value property*: Let Ω be a domain and $f \in A(\Omega)$. Suppose that $\mathbb{D}(z_0; \rho) \subset \Omega$. Then,

$$f(z_0) = \frac{1}{\pi \rho^2} \iint_{\mathbb{D}(z_0;\rho)} f(z) d\lambda(z).$$